Area Between Curves

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area between two curves**

Find the area of the region bounded by $y = x^2$ and $y = x + 2$.

Intersections: $x^2 = x + 2$, so $x^2 - x - 2 = 0$, $(x - 2)(x + 1) = 0$: $x = -1, 2$. Area $= \int_{-1}^{2} [(x + 2) - x^2]\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{2}$.

2 **[Foundational]** • **Target: Area between two curves**

The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.

$\int_0^4 \sqrt{x}\,dx = \left[\frac{2}{3}x^{3/2}\right]_0^4 = \frac{16}{3}$.

3 **[Foundational]** • **Target: Setting up with respect to y, and total area**

Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

Intersections: $y^2 = 2y$ gives $y = 0, 2$. Area $= \int_0^2 (2y - y^2)\,dy = \left[y^2 - \frac{y^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

4 **[Foundational]** • **Target: Setting up with respect to y, and total area**

Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

By symmetry, $\int_0^{\pi} \sin x\,dx = 2$ and $\left|\int_{\pi}^{2\pi} \sin x\,dx\right| = 2$. Total area $= 2 + 2 = 4$.

5 **[Foundational]** • **Target: More area problems**

Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).

Area $= \int_0^1 (x - x^3)\,dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

6 **[Foundational]** • **Target: Areas requiring case splits**

Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).

On $[-1, 0]$, $x^3 \geq x$; on $[0, 1]$, $x \geq x^3$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \int_0^1 (x - x^3)\,dx$. By symmetry both integrals equal $\frac{1}{4}$ (using the earlier result), so total area $= \frac{1}{2}$.

7 **[Foundational]** • **Target: Areas requiring case splits**

Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.

Intersections: $2x = x^2$ gives $x = 0, 2$. On $(0, 2)$, $2x > x^2$: Area $= \int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Find the area between $y = \cos x$ and $y = 0$ on $[0, \frac{\pi}{2}]$.

Area $= \int_0^{\pi/2} \cos x \, dx = [\sin x]_0^{\pi/2} = 1 - 0 = 1$.

9 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).

Area $= \int_0^1 (e^x - e^{-x})\, dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2$.

10 **[Intermediate]** • **Target: Visualizing a region bounded by two curves**

The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2)\, dx$.

The region is symmetric about $x = 0$ since both curves are even functions.
$2\int_0^2 (8 - 2x^2)\, dx = 2\left[8x - \frac{2x^3}{3}\right]_0^2 = 2\left(16 - \frac{16}{3}\right) = 2 \cdot \frac{32}{3} = \frac{64}{3}$, matching the earlier full-interval computation.

11 **[Intermediate]** • **Target: Area between a curve and a horizontal line**

Find the area between $y = 6 - x^2$ and $y = 2$.

Intersections: $6 - x^2 = 2$ gives $x = \pm 2$. Area $= \int_{-2}^{2} [(6 - x^2) - 2]\, dx = \int_{-2}^{2} (4 - x^2)\, dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = 2\left(8 - \frac{8}{3}\right) = \frac{32}{3}$.

12 **[Intermediate]** • **Target: Area between curves in applied contexts**

Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.

Intersection: $2t = t + 3$ gives $t = 3$. For $t < 3$, $v_2 > v_1$; for $t > 3$, $v_1 > v_2$. Gap $= \left|\int_0^3 (v_2 - v_1) dt\right| + \left|\int_3^{10} (v_1 - v_2) dt\right|$. $\int_0^3 (3 - t)\, dt = \left[3t - \frac{t^2}{2}\right]_0^3 = 9 - 4.5 = 4.5$.
$\int_3^{10} (t - 3)\, dt = \left[\frac{t^2}{2} - 3t\right]_3^{10} = (50 - 30) - (4.5 - 9) = 20 + 4.5 = 24.5$. Total gap $= 4.5 + 24.5 = 29$ m.

13 **[Intermediate]** • **Target: Area between curves in applied contexts**

Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).

$y = x^2 - 2x = x(x - 2)$ is ≤ 0 on $[0, 2]$. Area $= \left|\int_0^2 (x^2 - 2x)\, dx\right| = \left|\left[\frac{x^3}{3} - x^2\right]_0^2\right| = \left|\frac{8}{3} - 4\right| = \frac{4}{3}$.

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14 **[Intermediate]** • **Target: Area between curves in applied contexts**

Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.

On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \left|\int_{0}^{1} (x^3 - x)\,dx\right| = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} + \left|\left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{0}^{1}\right| = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Area between a curve and a horizontal line** • **Cross-Lesson: 3.01 The Derivative from First Principles**

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

1. Express x as a function of y: $y = \ln x \Rightarrow x = e^y$. 2. Apply the volume of revolution about the y-axis formula: $V = \pi\int_{c}^{d} [x(y)]^2\,dy = \pi\int_{0}^{1} (e^y)^2\,dy = \pi\int_{0}^{1} e^{2y}\,dy$. 3. Integrate: $V = \pi\left[\frac{e^{2y}}{2}\right]_{0}^{1} = \frac{\pi}{2}(e^2 - e^0) = \frac{\pi(e^2 - 1)}{2}$.

16 **[Challenging]** • **Target: Area between curves in applied contexts** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2\rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1\rangle$ are not scalar multiples ($1/2 eq -1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1 - 2) - \mathbf{j}(1 - 4) + \mathbf{k}(1 - (-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1).$ Magnitude.

17 **[Challenging]** • **Target: Area between two curves** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

Theorem (Area of Circle in Polar Coordinates): The curve $r = 2\cos\theta$ is a circle passing through the pole. Multiplying by r yields $r^2 = 2r\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x - 1)^2 + y^2 = 1$, which is a circle of radius $R = 1$ centered at $(1, 0)$. Step 1 · Identify integration bounds: The circle is completely traced as θ goes from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ (or from 0 to π). Step 2 · Set up and evaluate the polar integral.

18 **[Challenging]** • **Target: Setting up with respect to y, and total area** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

1. Normal vectors: $\mathbf{n}_1 = \langle 1, 1, 1\rangle$, $\mathbf{n}_2 = \langle 2, -1, 3\rangle$. 2. Direction vector $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$:

$\mathbf{d} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 1, 1; 2, -1, 3| = \mathbf{i}(3 - (-1)) - \mathbf{j}(3 - 2) + \mathbf{k}(-1 - 2) = (4; -1; -3).$ 3. Find a point on the line: set $z = 0$:

$\{x + y = 1; 2x - y = 4 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3},\ y = 1 - \frac{5}{3} = -\frac{2}{3}.$ P.

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19 [Challenging] • Target: More area problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

Differentiate implicitly with respect to x: $3x^2 - 2\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$
$3x^2 - 2y + (2y - 2x)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}$ A vertical tangent occurs when the denominator is zero (and numerator non-zero): $2y - 2x = 0 \Rightarrow y = x$ Substitute $y = x$ into the original curve equation:
$x^3 - 2x(x) + x^2 = 4 \Rightarrow x^3 - x^2 - 4 = 0$ Test integer factors: $x = 2$ gives $8 - 4 - 4 = 0$. Factoring gives $(x - 2)(x^2 + x + 2) = 0$.

20 [Challenging] • Target: Areas requiring case splits • Cross-Lesson: 6.01 Related Rates

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

A vertical tangent occurs when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} eq 0$. Compute $x'(t) = -\sin(t) = 0 \Rightarrow t = 0, \pi$. Check $y'(t) = 2\cos(2t)$: • At $t = 0$: $y'(0) = 2\cos(0) = 2 eq 0$. • At $t = \pi$: $y'(\pi) = 2\cos(2\pi) = 2 eq 0$. Therefore, vertical tangents occur at **$t = 0$** and **$t = \pi$**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Area between two curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$ ✓

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$
Step 2: Find the parameter k using the given total area of 8.
$\text{Area} = \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8:
$4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

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22 [Foundational] • Target: Setting up with respect to y, and total area

Parametric area: region bounded by $x(t), y(t)$ from $t = a$ to $t = b$: $A = \int y\,dx =$

a. $\int_a^b y(t)x'(t)dt$ ✓

b. $\int_a^b x(t)y'(t)dt$

c. $\int_a^b y(t)dt$

d. $\int_a^b x(t)dt$

$A = \int y\,dx = \int_a^b y(t)x'(t)dt.$

23 [Foundational] • Target: More area problems

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$ ✓

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

1. **Find the stationary point M:** Using the quotient rule to differentiate $y = \frac{\ln x}{x^2}$: $\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^2 - (\ln x)(2x)}{x^4} = \frac{x(1 - 2\ln x)}{x^4} = \frac{1 - 2\ln x}{x^3}$. Setting $\frac{dy}{dx} = 0$: $1 - 2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e}$. The y -coordinate at M is: $y = \frac{\ln(\sqrt{e})}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$. Since M is a stationary point, the tangent line L is horizontal wit.

24 [Foundational] • Target: Areas requiring case splits

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?

a. $\sqrt{2}$

b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$ ✓

d. 2

The area of the region is given by the definite integral of the function from $x = 0$ to $x = b$. The integral of $\frac{1}{x^2 + 1}$ is $\arctan(x)$. Therefore, the area is given by: $\text{Area} = \int_0^b \frac{1}{x^2+1} dx = [\arctan(x)]_0^b = \arctan(b) - \arctan(0) = \arctan(b)$ We are given that the area is $\frac{\pi}{3}$. So, we have: $\arctan(b) = \frac{\pi}{3}$ To find b, we take the tangent of both sides: $b = \tan\left(\frac{\pi}{3}\right)$ We know that.

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25 [Foundational] • Target: Areas involving trigonometric and exponential curves

Polar form of $y = c$ (horizontal line):

a. $r = c\csc\theta$

b. $r\cos\theta = c$

c. $r = c$

d. $r\sin\theta = c$ ✓

Since $y = r\sin\theta$ in polar coordinates, the horizontal line $y = c$ has polar equation $r\sin\theta = c$.

26 [Foundational] • Target: Visualizing a region bounded by two curves

Centroid $\bar{y}$ of the region bounded by $y = x^2$ and $y = 4$:

a. $\bar{y} = 5/2$

b. $\bar{y} = 8/5$

c. $\bar{y} = 2$

d. $\bar{y} = 12/5$ ✓

$A = 32/3$; $M_x = \int_{-2}^{2} \frac{1}{2}(16 - x^4)dx = 128/5$; $\bar{y} = (128/5)/(32/3) = 12/5$.

27 [Foundational] • Target: Area between a curve and a horizontal line

Volume of region bounded by $z = 1 - x^2 - y^2$ ($z \geq 0$):

a. $\pi/4$

b. $\pi/2$ ✓

c. π

d. $3\pi/4$

$V = \int_0^{2\pi}\int_0^1 (1 - r^2)r\,dr\,d\theta = 2\pi[r^2/2 - r^4/4]_0^1 = 2\pi(1/4) = \pi/2$.

28 [Foundational] • Target: Area between curves in applied contexts

Area of polar region swept by $r = \theta$ from 0 to 2π:

a. $\pi^3/3$

b. $4\pi^3/3$ ✓

c. $2\pi^3/3$

d. π^2

$A = \frac{1}{2}\int_0^{2\pi} \theta^2\,d\theta = \frac{(2\pi)^3}{6} = 4\pi^3/3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Area between two curves

Evaluate $\iint_R x\,dA$ where R is the region between $y = x^2$ and $y = x$.

a. $\frac{1}{6}$

b. $\frac{1}{24}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$ ✓

$\int_0^1 \int_{x^2}^{x} x\,dy\,dx = \frac{1}{12}$.

30 [Intermediate] • Target: Setting up with respect to y, and total area

The area of the region enclosed by $x = t - \sin t$, $y = 1 - \cos t$ (one arch of cycloid) using $A = \int y\,dx$:

a. πr^2

b. $3\pi r^2$ **(with $r = 1$)** ✓

c. π

d. 2π

One arch of cycloid: area $= 3\pi r^2$. With $r = 1$: 3π.

31 [Intermediate] • Target: More area problems

Line of intersection of two non-parallel planes: direction vector $=$

a. sum of normals

b. **cross product of normals** ✓

c. difference of normals

d. dot product of normals

Intersection direction $= \mathbf{n}_1 \times \mathbf{n}_2$.

32 [Intermediate] • Target: Areas requiring case splits

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

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33 [Intermediate] • Target: Areas involving trigonometric and exponential curves

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$ ✓

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ about the y-axis can be found using the shell method. The formula for the volume V is given by $V = \int_a^b 2\pi x f(x) dx$. In this problem, $f(x) = e^{-x^2}$, and the region is bounded by $y = e^{-x^2}$, the x-axis, and the line $x = a$ in the first quadrant. Thus, the limits of integration are from $x = 0$ to $x = a$. The volume of the solid is given by.

34 [Intermediate] • Target: Visualizing a region bounded by two curves

What is the area of the region bounded by the polar curve $r = \sqrt{5\sin\theta}$ for $0 \le \theta \le \pi$?

a. 5 ✓

b. 10

c. $\frac{5}{2}$

d. 5π

The area A of a region bounded by a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is given by the formula: $A = \frac{1}{2}\int_\alpha^\beta r^2\, d\theta$ Substitute $r = \sqrt{5\sin\theta}$, $\alpha = 0$, and $\beta = \pi$: $A = \frac{1}{2}\int_0^\pi (\sqrt{5\sin\theta})^2\, d\theta = \frac{1}{2}\int_0^\pi 5\sin\theta\, d\theta = \frac{5}{2}\int_0^\pi \sin\theta\, d\theta$ Evaluate the definite integral.

35 [Intermediate] • Target: Area between a curve and a horizontal line

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?

a. $\int_0^3 (x^2 + 2)\, dx$ ✓

b. $\int_0^3 (x^2 - 2)\, dx$

c. $\int_0^3 x^2\, dx$

d. $\int_0^3 2\, dx$

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\, dx$, provided that $f(x) \ge 0$ over the interval [a, b]. In this case, $f(x) = x^2 + 2$, which is always non-negative for all real x. The region is bounded by $x = 0$ and $x = 3$. Therefore, the area of the region is represented by the definite integral $\int_0^3 (x^2 + 2)\, dx$.

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36 **[Intermediate]** • **Target: Area between curves in applied contexts**

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?

a. 10 ✓

b. 12

c. 14

d. 16

The area of the region is given by the definite integral of $f(x)$ from $x = 1$ to $x = 2$. Area $= \int_1^2 (3x^2 + 1)\,dx$ First, find the antiderivative of $3x^2 + 1$: Antiderivative $= \frac{3x^3}{3} + x = x^3 + x$ Now, evaluate the definite integral using the Fundamental Theorem of Calculus: Area $= [x^3 + x]_1^2$ Area $= (2^3 + 2) - (1^3 + 1)$ Area $= (8 + 2) - (1 + 1)$ Area $= 10 - 2$ Area $= 8$

37 **[Intermediate]** • **Target: Area between two curves**

Green's first identity states (for smooth u, v on bounded Ω):

a. $\int_\Omega ablau \cdot ablav\,dV = \oint_{\partial\Omega} u \frac{\partial v}{\partial n} dS - \int_\Omega u\,\Delta v\,dV$ ✓

b. $\int_\Omega u\,\Delta v\,dV = \oint_{\partial\Omega} u \frac{\partial v}{\partial n} dS - \int_\Omega ablau \cdot ablav\,dV$

c. $\int_\Omega u\,\Delta v = \int_\Omega v\,\Delta u$

d. $\oint u\,\Delta v\,dS = \int_\Omega ablau \cdot ablav\,dV$

Green's first identity: $\int_\Omega ablau \cdot ablav\,dV = \oint_{\partial\Omega} u \frac{\partial v}{\partial n} dS - \int_\Omega u\Delta v\,dV$. Subtracting the symmetric version gives Green's second identity.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Setting up with respect to y, and total area** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

The total differential: $df = f_x\,dx + f_y\,dy + f_z\,dz$. For $f = xyz$, $df =$

a. $xyz(dx + dy + dz)$

b. $yz\,dx + xz\,dy + xy\,dz$ ✓

c. $y\,dx + z\,dy$

d. $x\,dx + y\,dy + z\,dz$

Applying Partial Derivatives and Gradient principles, $df = yz\,dx + xz\,dy + xy\,dz$. Therefore, the verified correct option is $yz\,dx + xz\,dy + xy\,dz$.

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39 [Challenging] • Target: More area problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Level curve: $f(x, y) = c$. The gradient $ablaf$ is $1cm$ to the level curve.

a. tangent

b. **perpendicular (normal)** ✓

c. parallel

d. zero

Applying Partial Derivatives and Gradient principles, $ablaf \perp$ level curve. Therefore, the verified correct option is perpendicular (normal).

40 [Challenging] • Target: Areas requiring case splits • Cross-Lesson: 6.01 Related Rates

Exponential growth model: if $\frac{dP}{dt} = kP$, then $P(t) =$

a. $P_0 + kt$

b. $P_0 kt$

c. ke^t

d. $P_0 e^{kt}$ ✓

Applying Transcendental Functions principles, Solution of $P' = kP$ is $P_0 e^{kt}$. Therefore, the verified correct option is $P_0 e^{kt}$.

41 [Challenging] • Target: Areas involving trigonometric and exponential curves • Cross-Lesson: 6.02 Optimization and Curve Sketching

Horizontal tangent to $y = x^3 - 3x$ at:

a. $x = 0$

b. $x = \pm 1$ ✓

c. $x = \pm\sqrt{3}$

d. $x = 3$

Applying Applications of Derivatives principles, $y' = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$. Therefore, the verified correct option is $x = \pm 1$.

42 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 7.01 Integration by Substitution

Cauchy MVT applied to f, g: gives $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$. It is used to derive:

a. Rolle's theorem

b. **L'Hôpital's rule** ✓

c. IVT

d. EVT

Applying Applications of Derivatives principles, Cauchy MVT → L'Hôpital. Therefore, the verified correct option is L'Hôpital's rule.

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43 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 7.02 Integration by Parts

Complex exponential: $e^{i\theta} =$

a. $\sin\theta + i\cos\theta$

b. $\cos\theta - i\sin\theta$

c. $\cosh\theta + i\sinh\theta$

d. $\cos\theta + i\sin\theta$ ✓

Applying Transcendental Functions principles, Euler's formula. Therefore, the verified correct option is $\cos\theta + i\sin\theta$.

44 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 8.02 Volumes of Revolution

Partial with respect to z: $f(x, y, z) = xyz^2$, $f_z =$

a. $2xyz$ ✓

b. xyz

c. yz^2

d. xy

Applying Partial Derivatives and Gradient principles, $\partial_z(xyz^2) = 2xyz$. Therefore, the verified correct option is $2xyz$.

45 [Challenging] • Target: Area between two curves • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Torsion $\tau = -d\mathbf{B}/ds \cdot \mathbf{N}$. For a plane curve, $\tau =$

a. 1

b. 0 ✓

c. κ

d. -1

Applying Vectors and Vector-Valued Functions principles, Plane curve: $\tau = 0$. Therefore, the verified correct option is 0.

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